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Deep Rice: A Comprehensive Deep Learning Approach for Quality Assessment in Rice Grains

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***Abstract*—Rice is a staple food and is one of the most extensively produced grain products worldwide. Ensuring that its quality is up-to-par is of utmost importance as it is heavily consumed all over the world. Rice quality is determined using morphological features, its texture, moisture content, and the nutritional profile of the grain. Finding the rice grain quality is extremely important as it plays an important role in identifying its consumer uptake, longevity, milling quality, and overall endorsement. This paper further delves into the classification of different grains and determining the grain quality. The types of rice grains used in this study are; Golden Ivy, Basmati, Parmal, Khandha Rice, and Sona Sella. The custom dataset created contains 3200 images for each variety of the regional rice of Uttarakhand,resulting in 16000 images in total. Various pre-processing techniques are applied to the images, through these images the morphological features are extracted, and then deep learning algorithms are applied to them to help classify the rice grains into their corresponding varieties. This results in an average validation accuracy of 93.96% over sixfold.**

Keywords—Rice, morphological, k-fold, keras, sklearn.

I. INTRODUCTION

Rice (Oryza sativa) is a primary food that is consumed all over the world, nourishing over half of its population. It is one of the most cost-effective nutrients and has the highest rates of both production and consumption. Since rice is mainly consumed as a whole grain, it makes assessing its quality is an essential and complex trait. Though, the quality of a grain can mean many things, in terms of rice it predominantly refers to its morphology, texture, cooking attributes, and nutritional profile. It is a perpetual challenge in the agro-food sector to ensure that the rice quality remains consistent. Hence, making the classification of rice is extremely important as it can help identify if different varieties of rice have been mixed, resulting in a decrease in the rice quality.

Traditionally, A human inspector performs the manual inspection of rice quality, albeit it can be laborious, yield inconsistent findings, and lack accuracy. However, advancements in technology have presented us with better opportunities to assess factors that influence rice quality. Deep learning and machine vision techniques offer solutions that support rice quality inspection. The main steps involved in the quality analysis of rice consist of; detection, segmentation, and classification. Various machine learning algorithms like random forest and logistic regression have been used to classify the rice. Artificial Neural Network, Convolutional Neural Network, and Deep Neural Network are also some algorithms that are used in this area.

This paper classifies the different rice varieties; Basmati, Golden Ivy, Khanda Rice, Sona Shaila, and Parmal using a dataset created using a good lighting system and a high-quality camera lens. Preprocessing techniques like erosion and dilation are applied to the dataset to enhance the images and morphological features like area, perimeter, major and minor axis length, aspect ratio, eccentricity, convex area, equivalent diameter, extent, solidity, roundness, compactness are extracted. Morphological features are the physical attributes of the grain and help to identify the differences between the rice varieties.The classification of rice grains in this paper employs a Deep Neural Network (DNN), a prominent algorithm in supervised learning renowned for its capacity to effectively capture intricate non-linear relationships. DNNs feature numerous hidden layers positioned between the input and output layers, facilitating comprehensive modeling of complex data patterns which progressively perform complex calculations on the input. DNN has a high accuracy score as each layer adds complexity to the model while enabling the model to process the inputs concisely for outputting the desired result. Hence, it can help provide optimal solutions for complex projects. Classification of rice can be successfully performed using deep neural networks, the result assisting in the quality analysis of the rice.

II. LITERATURE SURVEY

Table 1 gives a brief about the literature survey that was carried out.

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TABLE I. LITERATURE SURVEY

Authors	Methods Used	Practical Implications	Result
PanaddaKongsilp, Naratip Sangsai	The amalgamation of an image processing-based system with a deep learning methodology. Adoption of Mask-RCNN for detection, segmentation, and classification tasks	Automated, non-destructive analysis for rice quality evaluation. Cost-effective technique for rice quality classification. Capable of accommodating adjacent kernels and compact entities. Potential for meeting the Thai Agricultural Standard for rice quality.	The system achieves a mean average precision (mAP) of 0.96. The proposed system exhibits the ability and promise for classifying rice quality. It is adept at handling scenarios involving touching kernels and dense objects.
N. A. Kuchekar, V. Yerigeri	Digital Image Processing Techniques, Morphology-cal opening, Histogram equalization Wiener filter, Linear contrast adjustment, Gaussian filtering	Grading rice kernels based on morpho-logical techniques using computer vision methodologies. Preprocessing and segmentation of individual grains for classification. Extracting geometric features for grading based on grain length. Potential for future research on other quality features and moving images.	Assessment of parameters such as Major Axis Length and Minor Axis Length enables the quality analysis of Indian Basmati rice grains. Rice grains can be classified as Normal, Long, and Small based on calculated parameter values. Geometric features such as area, major axis length, and minor axis length are isolated and used for categorization. Rice grains are graded based on their length. The paper provides classification tables for different types of rice grains.
Murat Koku , Ilkay Cinar, Yavuz Selim Taspinar	Artificial Neural Network (ANN), Deep Neural Network (DNN), Convolutional Neural Network (CNN)	The study achieved high classification success rates for rice varieties. The CNN method showed the highest classification success rate of 100%. Misclassifications were observed due to the similarity between some rice varieties. The study can be used to develop effective and fast rice classification machines and also to identify authentic rice phenotypes	Classification successes: 99.87% for ANN, 99.95% for DNN, and 100% for CNN. Statistical results: sensitivity, specificity, pre-diction, F1 score, accuracy, false positive rate, and false negative rate were calculated. Confusion matrix: used to evaluate the performance of the classification algorithms. Cross-validation: Utilized to impartially assess the efficacy of model classification.
Bhupendra , Kriz Moses ,Ankur Miglani , Pavan Kumar Kankar	A machine vision framework comprising three stages: Image Acquisition, Preprocessing, and Classification, with CNN-based models employed for classification, thereby obviating the necessity for segmentation and feature extraction.	Development of a machine vision system for quality assessment of damaged rice grains. Application of Deep CNN models for accurate classification of damage in rice grains. Identification of the best-performing algorithm (EfficientNet-B0) for damage classification. Demonstration of high accuracy in classifying different types of damage in rice grains. Potential use of the classification tool for objective quality assessment of damaged rice grains.	EfficientNet-B0 is the best-performing model with 98.37% accuracy. Individual class accuracies range from 95.45% to 100%. EfficientNet-B0 has a model size of 47 MB and a prediction time of 0.122s.
Murat Koklu, Ilker Ali Ozkan	Image processing methods for noise removal and histogram equalization Otsu's method for thresholding the bean images. Extraction of dimensional and shape features from segmented bean images. Categorization of dried beans employing four distinct machine learning methodologies. Evaluation of model performance aimed at identifying the most effective approach.	Classification of dry bean seeds is important for seed uniformity and quality. The pro-posed computer vision system can successfully classify different types of dry beans. The model can be improved by using hybrid machine-learning methods and new algorithms. Including the third dimension of beans can increase the classification suc-cess. Texture and Incorporating statistical attributes can enhance the accuracy of classification outcomes. The MATLAB application can be turned into a mobile application for practical usage.	The study developed a computer vision system to classify seven different varieties of dry beans. The classification models (MLP, SVM, kNN, DT) achieved overall correct classification rates ranging from 87.92% to 93.13%. The SVM model had the highest accuracy of 93.13% and successfully classified all bean varieties. The precision values were proportional to the accuracy of the classifiers. The SVM model accurately classified Bar bunya, Bombay, Cali, Derma-son, Horoz, Seker, and Sira bean varieties with accuracy rates ranging from 86.84% to 100%. The SVM model had an accuracy of 93.19% in population planting with seven different dry bean seeds.

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III. METHODOLOGY

The methodology consists of four main steps; Data Acquisition, Feature Extraction, Model Building and Training, and Performance Evaluation. Figure 1 shows the proposed methodology.

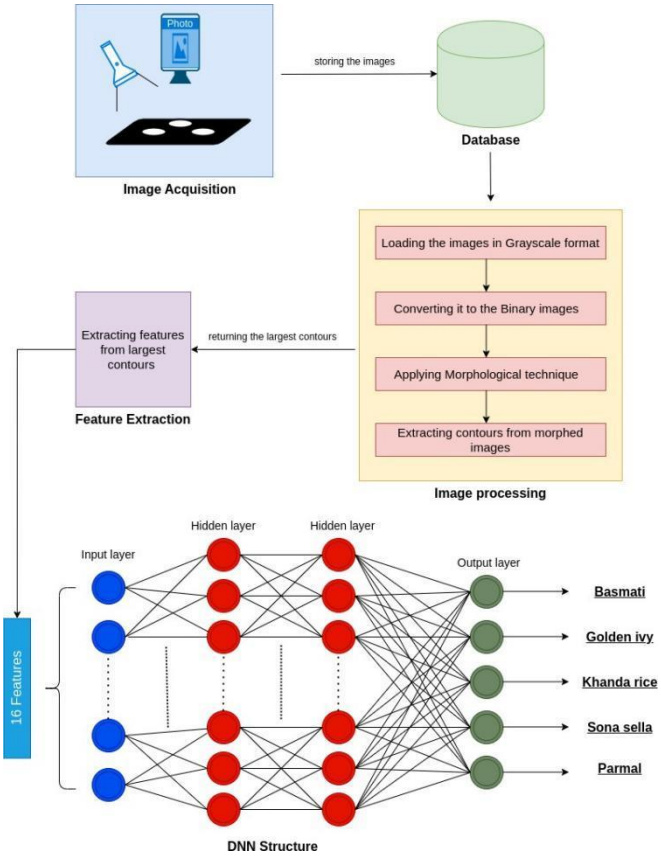

Fig. 1. Proposed Framework

A. Data Acquisition

In this process, we have focused on generating our image dataset using equipment like a camera, lighting system, and laptop. In consideration of the generation of our dataset, we visited the nearby general store to collect the different types of rice samples, we collected rice samples of 5 different varieties Basmati, Golden Ivy, Khanda rice, Sona Sella, and Parma. After collecting these samples, we arranged the equipment for clicking good-quality pictures of individual rice grains of each type. We have captured 3200 images of each of 5 varieties which is a total of 16000 images.

B. Feature Extraction

It is one of the most important steps to perform before training the deep learning model, but before this, we have applied image processing techniques to the dataset that we have generated. The captured images loaded to the device in grayscale format i.e. 8-bit color format vary from 0-255, where 255 stands for gray, 0 stands for black, and 127 stands for gray color. After that, Otsu's approach is used to threshold the grayscale images, with a threshold of 128. By creating a binary image where pixels with intensity values above the threshold are set to white (255) and those below are set to black (0), this binary thresholding technique seeks to distinguish the foreground (rice grains) from the background. The threshold image (thresh) is subjected to morphological procedures, namely dilation and erosion, to improve the grain contour. These procedures are performed with a kernel of size (5, 5). Using parameters for external contours and basic chain approximation, the cv2.findContours function extracts contours from the morphologically processed picture. The largest contour is chosen for further examination based on its area, with the assumption that it represents the rice grain. After contour selection, 12 geometric properties are extracted, such as Major Axis Length(L), Perimeter,Eccentricity, Area(A),Convex Area, Minor Axis Length(l), Aspect Ratio, Roundness, Equivalent Diameter, Extent, Solidity, Compactness. Furthermore, the four shape attributes and formulas that were obtained from these morphological characteristics, are given in Table II.

TABLE II. FEATURES AND FORMULA

Feature Name	Formula
Shape attribute 1	L/A
Shape attribute 2	L/A
Shape attribute 3	$A/((L/2)^2 * pie)$
Shape attribute 4	$A/((L/2) * (l/2) * pie)$

C. Model building and Training

Now that we have our dataset with a total 16 features extracted, Tensor Flow and Kera's to develop an automatic rice quality assessment model were used. Figure 1 shows the proposed framework for the same. Initially, the dataset is divided into training and validation subsets, enabling the training process and subsequent assessment of the Deep Neural Network (DNN). To accomplish this, the *sklearn. model selection* module's *train test split* method is used. The training set *(X train and y train)* comprises a majority *(80%)* of the dataset, while the remaining *20%* forms the validation set *(X val and y val)*. The division ensures that the model is trained on a sufficiently large dataset, while also providing an independent subset for assessing its generalization performance.

Rice grain morphology is complicated, but the DNN architecture is made to handle that. Using the Keras library, a sequential neural network architecture served as the basis for the endeavor to build a reliable model for the classification of tabular data. The model's sequential structure makes it possible for computations to go from input to output in an understandable, tiered fashion. An input layer built to absorb the dataset's features is where the model starts. There are 128 neurons in this layer, and they all use the Rectified Linear Unit (ReLU) activation function. ReLU was selected because of its capacity to introduce non-linearity, which enables the model to reflect intricate correlations found in the data.

Several hidden layers were added to the input characteristics

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to extract complex patterns and representations. These layers serve as intermediary processing units, gradually increasing the expressiveness of the data while decreasing its dimensionality. Three buried layers were identified: the first included 64 neurons, the second had 32 neurons, and the last one had 16 neurons. The ReLU activation function was utilized by every hidden layer to facilitate the acquisition of non-linear mappings. The output layer of the model is where the final predictions are generated. Neurons in this layer are arranged by the number of distinct classes found in the target variable. To normalize the output into a probability distribution across the classes and enable precise and assured predictions, the softmax activation function was selected.

After the architectural blueprint is finalized, the model needs to be assembled with instructions on how it is to be trained to learn and adapt. The model's convergence efficiency and speed are largely dependent on the optimizer selection. Because of its capacity for flexible learning rate, which makes it ideal for managing big datasets, the Adam optimizer has been chosen. Throughout training, Adam dynamically modifies the learning rate to help the model converge more quickly without going over the minimum. The selection of the loss function was guided by the nature of the classification assignment, which involved numerous classes. Sparse categorical cross-entropy was used, which is a good option for multi-class classification. This loss function quantifies the disparity between the actual class labels and the predicted class probabilities, providing the model with guidance to adjust its parameters. The accuracy metric was used to track the model's performance throughout training. As a key indicator of classification success, accuracy is the percentage of properly predicted cases.

The training phase began with the model architecture created and the compilation settings set. The model underwent train- ing for 50 epochs, signifying ten complete passes through the entirety of the training dataset. The choice of epochs balanced computational efficiency with the model's capacity to learn complex patterns. A batch size of 32 was selected, defining thequantity of samples processed before updating the model's parameters. This decision achieves a harmonious equilibrium between computational efficiency and gradient precision. To evaluate the model's generalization ability and prevent overfitting, a validation split of 20% was implemented.This reserved a portion of the training data for validation during each epoch, allowing the model to generalize to unseen instances.

D. Performance Evaluation

Following the model's training on the assigned training dataset, an independent, never-before-seen test dataset was used to thoroughly evaluate its performance.

This crucial assessment phase is necessary to determine how well the model generalizes and how well it predicts new, unobserved cases. A stratified6-fold Cross-Validation technique was utilized to evaluate the model's performance.This method involves dividing the dataset into K folds of equal size while ensuring that each fold retains the original dataset's class distribution. The model undergoes training on K-1 folds and validation on the remaining fold, iterated K times to ensure reliability.

IV. RESULT AND DISCUSSION

Preprocessing was done on the raw rice photos to improve clarity and lower noise as shown in. Important aspects of the rice grains were captured by extracting morphological features such as Area (A), Major Axis Length (L), Perimeter, Aspect Ratio, Minor Axis Length (l), Convex Area, Equivalent Diameter, Extent, Solidity, Roundness, Eccentricity, and Compactness. Table III shows a description of the rice dataset indicating several samples taken and the number of input features for each variety of rice.

TABLE III. DESCRIPTION OF RICE DATASET

Rice Variety name	Number of Samples	Number of Input features	Missing value
Basmati	3200	12	No
Golden Ivy	3200	12	No
Khanda rice	3200	12	No
sonasella	3200	12	No
Parmal	3200	12	No

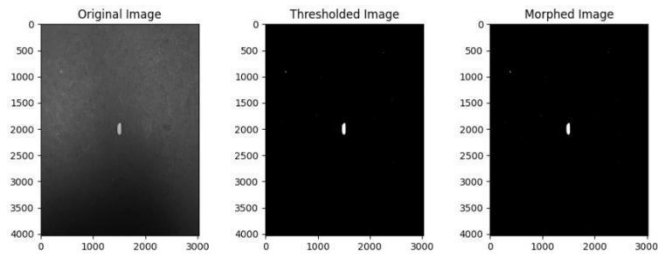

Fig. 2. Preprocessed Image

The experimental results demonstrate the effectiveness of the proposed deep neural network model for rice variety classification. The model achieved an average validation accuracy of 93.96% across 6 folds as shown in figure 3, indicating its ability to generalize well to unseen data. The Stratified K-Fold Cross-Validation approach helped assess the model's consistency and generalization performance across different subsets of the dataset.

As shown in Figure 4, evaluation metrics like f1 score, recall, and precision are calculated for each fold. As shown in Figure 5, the confusion matrix is also demonstrated for each fold.

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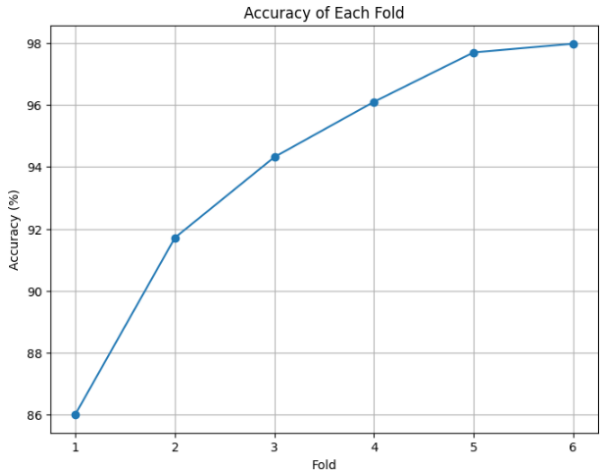

Fig. 3. Accuracy Graph of each fold

Fold	F1 Score	Recall	Precision
1	0.792022	0.795276	0.809209
2	0.867186	0.867642	0.86938
3	0.895209	0.895388	0.896017
4	0.934262	0.934383	0.9362
5	0.943716	0.943736	0.944447
6	0.941952	0.942236	0.94513

Fig. 4. Evaluation Metrics for each fold

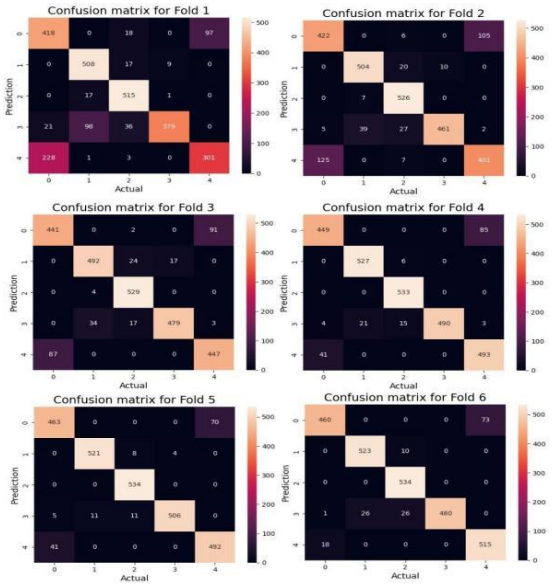

Fig. 5 Confusion matrix for each fold

V. CONCLUSION

In conclusion, the deep neural network model shows a great deal of promise for rice variety categorization, with applications ranging from crop management to food science and agriculture. Future research projects should put a high priority on improving and optimizing the model's scalability and performance. This means investigating cutting-edge methods to improve its design, including new features. And take advantage of developments in machine learning. For practical application and ongoing development, cooperation between researchers, industry stakeholders, and agricultural practitioners is crucial. This technology has the potential to transform rice categorization, promote sustainable agriculture methods, and improve global food security by utilizing the combined knowledge.

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